# CS506: Quantum Computing

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Topic: Grover's Algorithm – Search for an Item Among a List of Items

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## 1 Some Related Knowledge

1. Given $a_0, a_1, \ldots, a_{N-1}$ and query $y$, find $a_i$ such that $a_i = y$.

   **Sequential Search:**

   for $i = 0, 1, \ldots, N-1$:
   if $a_i = y$, print $a_i$

   The time complexity of sequential search is $O(N)$.

2. Reformulate the searching problem:

$$f : \{0, 1, \ldots, N-1\} \to \{0, 1\}$$

$$f(x) = \begin{cases} 1, & \text{if } y = a_x, \\ 0, & \text{if } y \neq a_x \end{cases}$$

   Given $f$, find $x$ such that $f(x) = 1$.

3. Encode

   Assume $N$ is a power of 2:

$$N = 2^n, \quad n = \log_2 N$$

   Then we can encode $0, 1, \ldots, N-1$ using $n = \log_2 N$ bits.

4. $U_f$ gate

$$U_f \left|x\right\rangle \left(\frac{\left|0\right\rangle - \left|1\right\rangle}{\sqrt{2}}\right) = (-1)^{f(x)} \left|x\right\rangle \left(\frac{\left|0\right\rangle - \left|1\right\rangle}{\sqrt{2}}\right).$$

   Thus, the action of $U_f$ on $\left|x\right\rangle$ can be represented by the phase oracle

$$O_f \left|x\right\rangle = (-1)^{f(x)} \left|x\right\rangle.$$

   In particular,

$$\text{if } f(x) = 0, \text{ then } O_f \left|x\right\rangle = \left|x\right\rangle,$$
$$\text{if } f(x) = 1, \text{ then } O_f \left|x\right\rangle = -\left|x\right\rangle.$$

   So the oracle flips the phase of the marked state.

# 2 Grover's Algorithm for Searching

Grover's algorithm uses

$$O(\sqrt{N})$$

flip gates and

$$O(\sqrt{N}\log_2 N)$$

other gates, with probability at least a positive constant. A natural question is: Can we improve this? For example, can we solve the search problem using $O(\sqrt[3]{N})$ flip gates? The answer is **No** according to a paper published in 1997.

## 2.1 SAT

Given a Boolean formula $\phi$ on $n$ Boolean variables

$$y_1, y_2, \ldots, y_n,$$

find an assignment of truth values to the variables that makes $\phi$ true (satisfied).

A naive brute-force method is to try all

$$2^n$$

possible combinations of variable values.

Define

$$f(x) = \begin{cases} 1, & \text{if } \phi \text{ is true by setting } y_1 = x_1, y_2 = x_2, \ldots, y_n = x_n, \\ 0, & \text{otherwise} \end{cases}$$

where

$$f : \{0, 1, \ldots, 2^n - 1\} \to \{0, 1\}.$$

Let

$$N = 2^n.$$

Then Grover's algorithm requires

$$O(\sqrt{N}) = O\left(2^{n/2}\right)$$

oracle gates.

## 2.2 Quantum Circuit for Grover's Algorithm

Shown in Figure 1.

## 2.3 State Evolution in Grover's Algorithm

Initially, the system is in the state

$$|\psi_0\rangle = |0\rangle|0\rangle|0\rangle \cdots |0\rangle = |0\rangle^{\otimes n},$$

where there are $n$ qubits.

Applying the Hadamard gate to each qubit gives

$$|\psi_1\rangle = H|0\rangle\, H|0\rangle \cdots H|0\rangle = H^{\otimes n}|0\rangle^{\otimes n}.$$

Since $N = 2^n$, this can be written as

$$|\psi_1\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle.$$

After applying the oracle $O_f$, we obtain

$$|\psi_2\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} O_f|x\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)}|x\rangle.$$

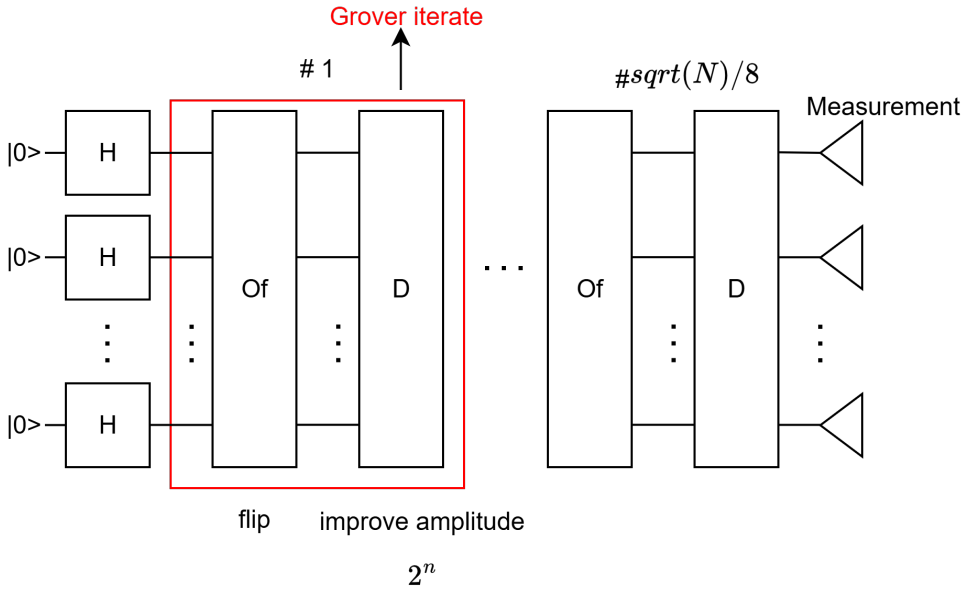


Figure 1: Quantum circuit for Grover's algorithm. The circuit begins with Hadamard gates to create an equal superposition, followed by repeated Grover iterations consisting of the oracle $O_f$ and the diffusion operator $D$, and ends with measurement.

Suppose $x^*$ is the marked state. Then

$$f(x^*) = 1, \qquad f(x) = 0 \quad \text{for } x \neq x^*.$$

Therefore,

$$|\psi_2\rangle = -\frac{1}{\sqrt{N}}|x^*\rangle + \frac{1}{\sqrt{N}} \sum_{\substack{x \in \{0,1\}^n \\ x \neq x^*}} |x\rangle.$$

Thus, the oracle flips the phase of the marked state.